ANALYSIS I — SHEET 4 (for Week 6 tutorials)
Algebra of Limits. Monotone Sequences. Base Expansions

1. For which of the following a_n does the sequence (a_n) converge? Justify your answers, and find the value of the limit when it exists.

$$\text{(i)}\quad \frac{n^2+n+1}{n+1}; \qquad \text{(ii)}\quad \frac{n+1}{n^2+n+1}; \qquad \text{(iii)}\quad \frac{3n^3+n^2\sin n}{2n^3-100n-3};$$

$$\text{(iv)}\quad \frac{2^n n^2+3^n n}{3^n(n+1)+n^7}; \qquad \text{(v)}\quad \sqrt{n+1}-\sqrt{n}; \qquad \text{(vi)}\quad \frac{(n!)^2}{(2n)!}.$$

[You may quote standard properties of the trigonometric functions even though these have yet to be proven in the course.]

2. By considering the intervals $[(3k-1)\pi/3, (3k+1)\,\pi/3]$, where $k \geqslant 1$, show that the sequence $\cos n$ diverges.

3. Let a_n be a sequence of non-negative terms such that $a_n \to L$. Show that $L \geqslant 0$ and that $\sqrt{a_n} \to \sqrt{L}$.

4. Given any $a_1 > 0$, define a sequence (a_n) by

$$a_{n+1} = \frac{(a_n)^2+3}{2a_n} \quad \text{for } n \geqslant 1.$$

(i) Prove that $a_n \geqslant \sqrt{3}$ for all $n \geqslant 2$.
(ii) Prove that $a_{n+1} \leqslant a_n$ for all $n \geqslant 2$.
(iii) Prove that (a_n) converges and find the limit of a_n as $n \to \infty$.
(iv) On the same axes, sketch $y = (x^2+3)\,/\,(2x)$ and $y = x$ and, via cobwebbing, the behaviour of (a_n).

5. Show that the sequence of sums $\sum_{k=0}^{n} \frac{1}{k!}$ is increasing and bounded above. *[Hint: compare these sums with a geometric sum.]* The limit of this sequence is denoted by e. Show, for $n \geqslant 1$, that

$$0 < e - \sum_{k=0}^{n} \frac{1}{k!} < \frac{1}{n(n!)}.$$

Deduce that e is irrational.

6. An equivalent formulation of the completeness axiom is the following: *bounded monotone sequences converge.* This exercise shows the equivalence of this axiom with our original completeness axiom, namely the existence of suprema.

Let S be a non-empty bounded set. We now define two sequences (s_n) and (u_n) as follows. Take $s_1 \in S$ and let u_1 be an upper bound for S. Assuming $s_1, \ldots, s_n$ and $u_1, \ldots, u_n$ have been defined,

- if $\frac{1}{2}(s_n+u_n)$ is an upper bound for S, set $u_{n+1} = \frac{1}{2}(s_n+u_n)$ and $s_{n+1} = s_n$;
- if $\frac{1}{2}(s_n+u_n)$ is not an upper bound for S, set $u_{n+1} = u_n$ and choose s_{n+1} to be an element of S greater than $\frac{1}{2}(s_n+u_n)$.

(i) Show that (s_n) is increasing and bounded. By assumption, (s_n) converges and set $\alpha = \lim s_n$.
(ii) Show that α is an upper bound for S.
(iii) Show that $\alpha = \sup S$.

Extension Exercises (Optional)

7. A unique *binary* (base 2) expansion of a real number may be defined in the same way decimal expansions were defined in lectures. The first digit to the right of the binary point denotes $\frac{1}{2}$s, the next digit $\frac{1}{4}$s, the next $\frac{1}{8}$s, etc.

(i) Show that $0.101101101101\ldots$ is the binary expansion of $5/7$.
(ii) Find the binary expansion of $1/9$.

8. Let (x_n) be a bounded real sequence and define

$$s_n = \sup\{x_m \mid m \geqslant n\}, \qquad i_n = \inf\{x_m \mid m \geqslant n\}.$$

(i) Show that s_n is a bounded decreasing sequence and i_n is a bounded increasing sequence. Hence (s_n) and (i_n) each converge and we define

$$\limsup x_n = \lim s_n, \qquad \liminf x_n = \lim i_n.$$

These limits are respectively known as the *limit superior* and the *limit inferior*.
(ii) Show that $\liminf x_n \leqslant \limsup x_n$.
(iii) For bounded real sequences (x_n) and (y_n), is it true that

$$\limsup (x_n + y_n) = \limsup x_n + \limsup y_n?$$

(iv) Find the limit superior and limit inferior of the following sequences.

$$\text{(a)} \quad (-1)^n, \qquad \text{(b)} \quad \frac{(-1)^n}{n}, \qquad \text{(c)} \quad \frac{(-1)^n\, n + 1}{n + (-1)^{n+1}} \qquad \text{(d)} \quad \sqrt{n} - \lfloor \sqrt{n} \rfloor.$$

9.(i) If (x_n) is convergent, show that $\liminf x_n = \limsup x_n = \lim x_n$.
(ii) If $\liminf x_n = \limsup x_n$ show that (x_n) converges.

10. (i) For $n \geqslant 1$, set

$$I_n(x) = x^{2n+1} \int_{-1}^{1} (1 - t^2)^n \cos(xt)\, \mathrm{d}t.$$

Use integration by parts to show that

$$I_n(x) = 2n(2n-1)I_{n-1}(x) - 4n(n-1)x^2 I_{n-2}(x).$$

(ii) Show that $I_0(x) = 2\sin x$ and $I_1(x) = -4x\cos x + 4\sin x$. Deduce that there are polynomials $P_n(x)$ and $Q_n(x)$, of degree at most $2n$ and with integer coefficients, such that

$$I_n(x) = n!(P_n(x)\sin x + Q_n(x)\cos x).$$

(iii) Let $x = \pi/2$ and suppose for a contradiction that $\pi/2 = a/b$ where a and b are positive integers. Show that

$$\frac{a^{2n+1}}{n!} \int_{-1}^{1} \left(1 - t^2\right)^n \cos(\pi t/2)\, \mathrm{d}t$$

is an integer for all n. Explain how this leads to a contradiction and conclude that π is irrational.

11. (i) Use calculus to show for $x > 0$ that

$$\sin x \leqslant x \qquad \text{and} \qquad \cos x \geqslant 1 - \frac{x^2}{2}.$$

(ii) Use the irrationality of π and Dirichlet's approximation theorem (Sheet 3, Exercise 9) to show that $\limsup \cos n = 1$.